# Requirement Analysis Phase Template

## Technology Stack.

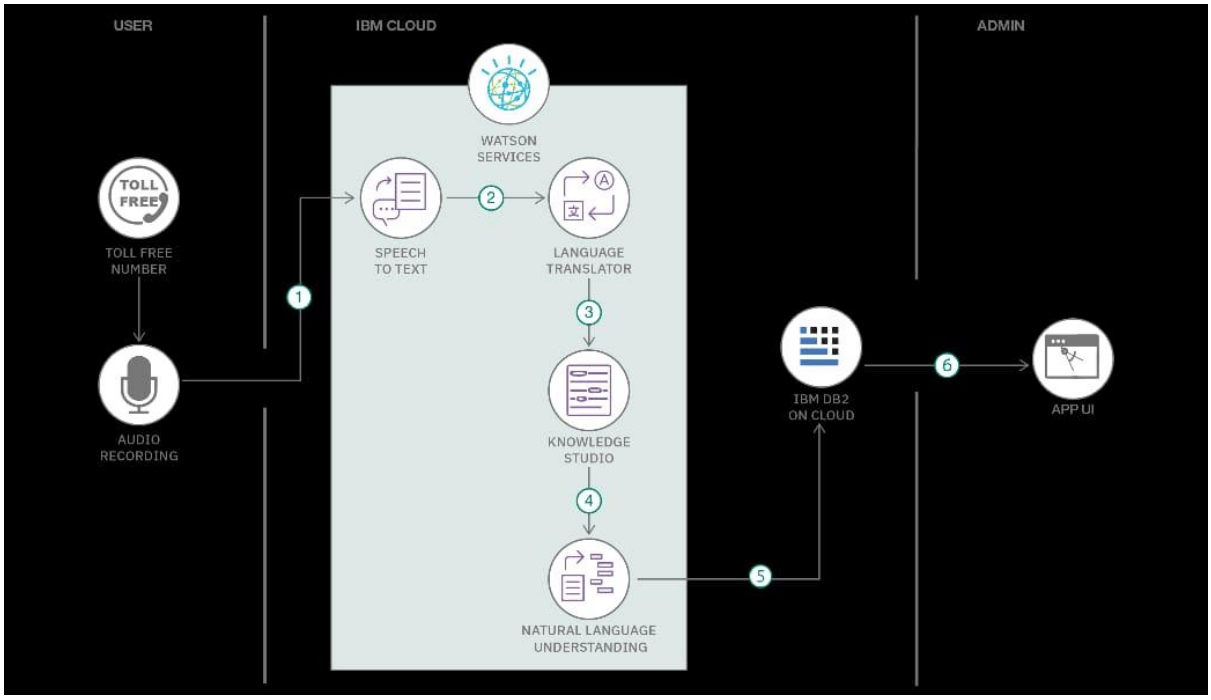
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|---|---|
| **DATE:** | **27/09/2026** |
| **TEAM ID:** | **SWTID-2026-2642** |
| **PROJECT NAME:** | **Weather wise** |
| **MAXIMUM MARKS:** | **4 Marks** |

### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



**Guidelines:**

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.) Indicate Data Storage components / services Indicate interface to machine learning models (if applicable)

**Table 1: Components & Technologies.**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | User Interface | User registration, login, AI-FITTRACK search and AI Library answers. | HTML, JavaScript |
| 2 | Application Logic-1 | User registration, login and authentication. | Node.js, Express.js, JWT, bcryptjs |
| 3 | Application Logic-2 | Create, view, update and delete AI-FITTRACKs. | Node.js, Express.js |
| 4 | Application Logic-3 | Generate Al answers and topic suggestions. | Google Gemini API |
| 5 | Database | Store user details and AI-FITTRACK records. | MongoDB, Mongoose |
| 6 | Cloud Database | Store application data online, if used. | MongoDB Atlas |

## Table 2: Application Characteristics

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1 | Open-Source Frameworks | Uses open-source frameworks for frontend, backend and database operations. | Express.js, Node.js, MongoDB |
| 2 | Security Implementations | Uses password hashing, JWT authentication and protected routes. | bcryptjs, JWT, HTTP |
| 3 | Scalable Architecture | Modular REST API architecture supports adding more users and AI-FITTRACKs. | Node.js, Express.js, MongoDB |
| 4 | Availability | Provides access to published AI-FITTRACKs and AI-assisted answers through the application. | Node.js Server / Cloud Hosting |
| 5 | Performance | Uses database queries and semantic search to retrieve relevant AI-FITTRACKs and generate answers. | MongoDB, Mongoose, Google Gemini API |